Service Bulletin

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Cummins Inc. Inspection of Diesel Engines Subjected to Flooding Conditions	

Cummins Inc. Inspection of Diesel Engines Subjected to Flooding Conditions

This document provides a process that can be followed to inspect Cummins® diesel engines that have been partially or totally submerged in water. The process outlined in this document can help to prevent engine damage when placing the engine back into service.

Note : The process outlined in this document nor the use of the process carries any express or implied warranty by Cummins Inc. The process outlined in this document does not completely eliminate the possibility of a malfunction that is the result of the engine being partially or totally submerged during a flood. Damage to the engine as the result of being in a flood or an engine malfunction that is the result of that damage is not a material malfunction or workmanship on the part of Cummins Inc. and will not be covered by Cummins® warranty.

The degree of engine damage caused by submersion under water is driven by the following factors:

- How long the engine was submerged under water.
- The content of the water or liquid (fresh, salt, brackish, acidic, and so forth).
- The length of time and the ambient humidity from the time the engine is removed from submersion in water to being put back in to service.
- If the engine was running when submerged under water.
- The condition of the engine before submersion in water.

The process outlined in this document assumes the following:

- One does **not** known if and/or how long an engine has been totally or partially submerged in water.
- The engine was **not** running when the engine became submerged.

Inspection Steps

Note : This process assumes one does not know if and/or how long an engine has been totally or partially submerged in water, and that the engine was not running when submerged.

⚠ CAUTION ⚠

Do not attempt to start the engine until the process outlined in this document has been completed. Any attempt to start the engine when there is water inside the cylinder will cause serious damage to internal engine components.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

Preparatory Steps

- Disconnect the machine batteries. See equipment manufacturer service information.
- Drain 1 quart to 1 gallon of lubricating oil from the lubricating oil pan.

Is there water in the lubricating oil?

- Yes:
  - Change the lubricating oil and lubricating oil filters.
  - Continue with No.
- No:
  - Drain 1 quart to 1 gallon of fuel from the drain plug on all fuel tanks.

Is there water in the fuel?

- Yes:
  - Drain and replace all of the fuel in the tanks, fuel hoses and the fuel filters. Clean the interior of the fuel tank and hoses.
  - Drain some fuel from the fuel pump and check for water. If water is present in the fuel, completely drain the fuel pump and fill with clean fuel.
  - Continue with No.
- No:
  - Attempt to rotate the crankshaft by hand using **only** hand tools. Do **not** crank the engine with the starting motor.

Will the crankshaft turn without excessive resistance for two full crankshaft revolutions?

- No:
  - Remove all of the injectors. Disconnect the air compressor air outlet hose. Rotate the crankshaft to top dead center (TDC) for each cylinder and remove any liquid from the cylinder. Squirt 1/2 to 1-ounce of clean engine lubricating oil into each cylinder and rotate the engine no less than two revolutions. Replace or rebuild the injectors. Remove any water found under the rocker lever covers. Make note of the amount of corrosion/rust in this area.
  - Continue with Yes.
- Yes:
  - Remove all air filters and check air filters and air filter housings for any sign of water.

Is there any indication that the filter element(s) have been or are wet?

- Yes:
  - Replace all air filters. Remove all air intake piping and clean any rust or corrosion. Check the compressor side of the turbocharger for any sign of debris, rust or corrosion. Rebuild or replace the turbocharger if debris, rust, or corrosion are found. All salt or brackish water **must** be flushed from any aluminum or aluminized steel tubing to prevent corrosion and entry of aluminum oxide into the power cylinder. Remove and clean the charge air cooler or aftercooler, if equipped.
  - Continue with No.
- No:
  - Drain 1 to 4 liters [1 to 4 quarts] of diesel exhaust fluid (DEF) from DEF tank. Inspect quality of DEF sample using refractometer, Part Number 4919554.

Is the quality of the DEF out of specification?

- Yes:
  - Drain DEF tank, DEF hoses, and DEF Filters.
  - Clean interior of DEF tank, flush hoses, replace DEF in-tank filter and tank vent, refill tank with clean DEF, and replace aftertreatment DEF dosing valve.
  - Drain some DEF from aftertreatment DEF dosing unit and check quality with refractometer. If quality of DEF is out of specification, completely drain aftertreatment DEF dosing unit , flush with clean distilled water, and fill with clean DEF. Continue with No.
- No:
  - Remove diesel particular filter (DPF), diesel oxidation catalyst (DOC), and selective catalytic reduction (SCR) from unit.

Was the aftertreatment DOC, DPF, SCR, or tailpipe outlet submerged in water?

- Yes:
  - Orient the DPF, DOC, and SCR with inlet facing down to drain water for 10 minutes. Orient the DPF, DOC, and SCR with outlet facing down to drain water for 10 minutes.
  - Continue with No.
- No:
  - Remove exhaust pipe between turbocharger and muffler/converter.

Is there or has there been any water in the turbocharger, pipe or muffler/converter?

- Yes:
  - Inspect the turbocharge turbine wheel and turbine housing for debris or corrosion. Rebuild or replace the turbocharger if debris or corrosion are found.
  - Remove and clean the exhaust gas recirculation (EGR) cooler. It is **not** necessary to pressure test the EGR cooler.
  - Continue with No.
- No:
  - Starting motors and alternators can be flushed to remove salt or debris. Use a cleaner intended for electric motors. These components are very difficult to clean and can suffer premature malfunction as the result of immersion in water.
  - If the engine has been subjected to a flood in salt or brackish water, steam clean the entire engine exterior to remove the salt.

- Start the engine and listen for any unusual noises. If there are unusual noises, see the corresponding Troubleshooting and Repair Manual or Service Manual. Reference the applicable Engine Noise Excessive troubleshooting.
- There can be some white smoke from lubricating oil sprayed into the power cylinder, but the smoke should clear in 5 to 10 minutes.
- If the engine will **not** start or operate correctly, see the corresponding Troubleshooting and Repair Manual or Service Manual. Reference the applicable Engine Difficult to Start or Will **Not** Start troubleshooting.
- Troubleshoot any fault codes that can appear on the INSITE™ electronic service tool.
- Operate the engine under load as soon as possible to remove any residual internal moisture.
- Change the lubricating oil and lubricating oil filters. See the recommended maintenance schedule found in the operation and maintenance manual.

Document History

Date	Details
xxx	Module Created
2012-12-3	QSOL Quick Fix|Reason: Spelling Error|Notes: none
2016-9-12	Added DEF quality check and DEF out of specification section. Added DOC, DPF, SCR, or tailpipe outlet submerged in water section. Modified water in the turbocharger, pipe or muffler/converter section.

Last Modified: 06-Oct-2016